Module 11

Client Intake

The client intake is the most critical element of the training process, as it determines if a client is healthy enough to participate in a running program. Additionally it establishes a starting benchmark in regard to various training elements.

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Elements of the client screening
- Anthropometric assessments
    - Girth measurements
    - Body composition
- Why BMI and BIA are not advised for assessments
- Client profile
- Program design variables
- Working with special populations

❖ CLIENT SCREENING

It is mandatory that all clients are considered apparently healthy prior to receiving coaching. This is accomplished through the use of the HHQ and PARQ, as detailed below. Additionally, the *client profile* will help you better understand your client to ensure the most efficient coaching/client relationship.

CLIENT HEALTH SCREENING

Before you coach any client, two forms will need to be filled out:

1. **Physical Activity Readiness Questionnaire (PARQ)**
2. **Health History Questionnaire (HHQ)**

The gold standard for the PARQ is the *American College of Sports Medicine (ACSM),* which is reprinted from the *Canadian Society for Exercise Physiology* (497).

> As laws vary from state to state, it is advised to seek legal advice prior to coaching individuals utilizing the PARQ and HHQ.

Physical Activity Readiness Questionnaire (PARQ)

Based on ACSM protocol, if a client answers **yes** to any of the PARQ questions, the person must obtain written consent from a physician providing clearance to work out before coaching can begin. However, based on UESCA protocol, any client, regardless of PARQ or HHQ results, *must* have been cleared by a physician in the last 12 months prior to beginning a running program. The questions on the PARQ are (497):

- Has your doctor ever said you have a heart condition and recommended medically supervised physical activity?
- Do you have chest pain brought on by physical activity?
- Have you developed chest pain within the last month?
- Do you lose consciousness or fall over as a result of dizziness?
- Do you have a bone or joint problem that could be aggravated by physical activity?
- Has a doctor ever recommended medication for high blood pressure or a heart condition?
- Are you aware of any reason your doctor would advise you against physical activity without medical supervision?

Even if cleared, a physician may suggest certain guidelines such as keeping a client's heart rate below a certain limit or avoiding particular exercises. Regardless of your opinion of what a physician says, the doctor's "prescription" for your client supersedes everything and must be followed exactly as intended.

When administering the PARQ, it is advised to ask clients the questions versus have them fill in the answers themselves. If you are working with a client remotely, schedule a call to go over the PARQ instead of emailing the form.

Health History Questionnaire (HHQ)

The HHQ is important from both a liability and performance aspect. For example, past or present injuries of a client can affect biomechanical efficiency and overall performance. Medications that your client takes can also affect performance. This is especially true regarding the cardiovascular system. Common types of medications prescribed that affect heart rate are:

- Beta blockers
- Antidepressants
- Vasodilators
- Bronchodilators
- Calcium channel blockers
- Nitrates

If your client does not know what type or classification a drug is, it is the person's responsibility to contact a physician to understand what, if any, effects it can have on the body. Additionally, it is your responsibility to be aware of the effects that a drug can have on a client, and modify the program accordingly. For example, beta blockers are used for chest pain as well as high blood pressure and can lower one's heart rate. After a physician has cleared a client to exercise, you must modify the program to take into account any effect a medication has on the body (i.e., lowered heart rate). For clients with asthma, it is strongly recommended that they always have a fast-acting inhaler on their person in the event of an acute asthma attack.

While there are various HHQ forms and questions, below are common questions found on HHQ forms:

1. Current Height:
2. Current Weight:
3. Age:
4. Sex:
5. What is your resting heart rate upon waking:

6. Do you take any drugs or medications?
7. Date of your last medical physical:
8. Do you have a family history of heart disease?
9. Do you have emphysema?
10. Are you, or have you been, recently pregnant?
11. Do you have high cholesterol?
    a. If so, are you under the care of your physician for it?
12. Have you had surgery in the past year?
13. Have you ever had an injury that caused you to stop exercising for more than a week?
14. Are you, or have you ever been, anorexic or bulimic?
15. Are there any other physical or emotional problems that may affect your training?
16. Do you drink alcohol?
    a. If so, how much per week?
17. Do you have any metabolic diseases such as diabetes, hyperthyroidism, etc.?
18. Do you, or have you ever, smoked regularly?
19. Do you currently or have you had any of the following conditions?
    - Heart Disease
    - Heart Attack
    - Heart Surgery
    - Heart Murmur
    - Hypertension
    - Thyroid Problems
    - Asthma
    - Epilepsy
    - Anemia
    - Stress Fracture

As noted previously, it is UESCA protocol for your clients to be cleared within the last 12 months by their physician to train and participate in a coached running program. However, if your client answered **yes** to any of questions 8, 9, 10, 11, 12, 13, 14, 15, 18, and 19 in the preceding list, it is advised to have the person revisit a doctor to ensure that participation in a running program is medically approved.

Cardiac Screening

It is advised that your client receive a cardiac assessment (EKG stress test) within one year prior of beginning a running program. A successful stress test does not rule out the chance of having a cardiac event, nor does it identify all preexisting cardiac issues. However, this step ensures that clients have done everything in their power to ensure that they are "apparently healthy" from a cardiovascular perspective prior to beginning a training program.

Questions/Forms

As noted previously, it is strongly advised that the PARQ and HHQ questions noted in this section are not to be used "as is" without consulting legal counsel.

ANTHROPOMETRIC ASSESSMENTS

Anthropometry relates to the measurements and proportions of the human body. The anthropometric measurements listed next represent the most common and relevant measurements relative to fitness and sports performance. The true value of anthropometric measurements is gauged through the correlation of initial assessments and reassessments of tests.

Anthropometric assessments are commonly performed at the beginning, end, and throughout a training program. These assessments are most commonly done for clients who have a focus on weight loss.

The opportunity for error is greatly reduced by having the same person perform the initial assessments and reassessments.

GIRTH MEASUREMENTS

Girth measurements are circumference measurements taken at various locations on the body. The two most important factors in obtaining accurate measurements are the location and tension of the tape measure. When retesting, the exact location and tape tension must be replicated. While it is always best that reassessment is performed by the same individual who took the initial assessment, the testing apparatus plays a role in accuracy as well. The *Gulick tape measure* is made for the purpose of girth measurements. It has a small spring at the end of the tape that expands after the tape becomes tight. Upon the

initial stretch of the spring, the measurement is taken. This ensures that the correct amount of tension is on the tape, as it does not allow for overtightening.
While the locations differ based on what areas you are looking to assess, the most common areas are:

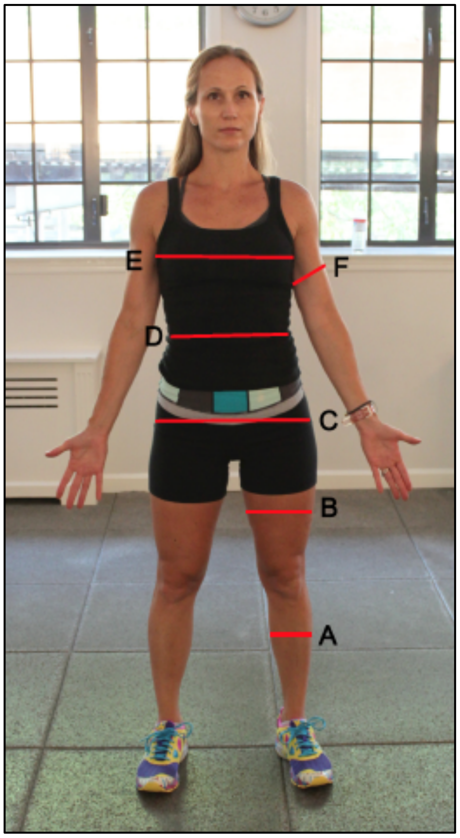

Figure 11.1 Girth measurements sites

A: Calves: Measure at the widest part of the calves.

B: Thigh: Measure the middle section of the thigh. To get an accurate site measurement, have clients lift their leg until their femur is parallel to the ground. Then have them place a finger at the crease formed by the hip angle, called the *inguinal crease.* The midpoint between the inguinal crease and the top of the patella (kneecap) is the location of the testing site.

C: Hips: Measure at the widest part of the hips and take the measurement from the side of your client.

D: Abdomen: Measure level with the *navel* (belly button). As with the chest measurement, if you are a male assessing a female, have the client adjust the tape in front and take the measurement on the side or behind your client.

E: Chest: Measure level with the nipples, while making sure the tape is horizontal with the ground. If you are a male assessing a female, have the client adjust the tape in front and take the measurement from behind.

F: Arm: Measure at the widest part of the biceps (unflexed).

BODY COMPOSITION

The two most prevalent methods for assessing body composition are:

1. Skinfold Calipers
2. Bioelectrical Impedance

Skinfold Calipers

As the name suggests, skinfold calipers take measurements of folds of skin. There are many different locations on the body to measure based on the particular formula or equation being used. The measurements are assessed in millimeter increments. Skinfold calipers are typically spring-loaded with a preset amount of tension. The spring tension is important because if the tension is too high or too low, the compression on the skinfolds will be inaccurate and the percent of body fat will be wrong.

Excessive body fat and tester error can adversely affect the accuracy of the test. The more body fat an individual has, the higher the percent error will typically be.

If the tester does not administer the test properly and places the calipers in the wrong location or pinches the skin with an incorrect amount of pressure, the test result will be inaccurate. UESCA references the *Jackson-Pollock 3-Site Test* (498).

Administering the Assessment

For this test, you will need to know your client's age and body weight. After you get these pieces of information, you will want to inform your client that the measurements will pinch slightly and that you will assess each site three times and take the average. Perform this test in a private setting to make the client feel more comfortable.

Pinch the fat and skin in the designated area and place the calipers right next to your pinch. Once you have placed the calipers in the right place, allow the calipers to "settle in," which takes about two or three seconds. Note the measurement then open the calipers and remove them from the site that was tested. Pulling the calipers off the site without opening them will pinch the client.

Prior to measuring a site, instruct your client not to tighten the muscle at the site being tested. When performing a pinch, you want to make sure you pinch only the skin and fat, not muscle. There are two ways to do this:

1. After making the pinch and while holding it, gently pull away from the surface of the body part being assessed. If a muscle is pinched, it will likely "slip" out of the pinch, leaving only the skin and fat.

2. After you have pinched the site and before you apply the calipers, ask the client to tighten the muscle in the area being tested. If you feel the area you pinched tighten or slide out from the pinch, you pinched muscle.

You should perform the pinch using your thumb and side of the forefinger (pointer finger). If you have long fingernails, use the sides of both fingers so you do not scratch your client.

Because of the tautness of the skin, the thigh measurement is often the most difficult one to perform accurately. To make this site easier to measure, have a client shift his or her weight onto the leg not being measured. The leg being measured should have a slight bend in the knee with all the body weight on the opposing leg.

All sites are on the right side of the body.

Female Measurement Sites

1. **Tricep:** (vertical pinch) In the middle of the upper arm, midway between the elbow and the top of the shoulder (*acromion process*).

Figure 11.2 Tricep caliper site

2. **Suprailiac Crest**: (diagonal pinch) Have the client find the top of the iliac crest by moving the thumb down the right side (*midaxillary line*) of the body until hitting the top of the pelvic bone. This area is called the *iliac crest*. Once the client has located the iliac crest, diagonally pinch the skin/fat directly in front of it.

Figure 11.3 Suprailiac caliper site

3. **Thigh**: (vertical pinch) Same location as the thigh girth measurement noted previously, midway between the *inguinal crease* and the top of the *patella*.

Figure 11.4 Thigh caliper site

Male Measurement Sites

1. **Chest:** (diagonal pinch) Halfway between the nipple and armpit (*axillary fold*).

Figure 11.5 Chest caliper site

2. **Abdomen:** (vertical pinch) Approximately one inch laterally to the right of the belly button (*umbilicus*).

11.6 Abdominal caliper site

3. **Thigh:** (Vertical Pinch) Same location as the female thigh girth measurement noted above.

Body Fat Equation

To get one's body fat percentage, *body density* must first be determined. The Jackson-Pollock formula is used for determining body density (85, 86). Once body density is determined, through use of a formula developed by W.E. Siri, one's body fat percentage can be ascertained (84). While the body-density equation differs between male and female, the body-fat equation (Siri) is the same for both genders.

Because of the labor-intensive nature of the body-density equations, use the following link (body-composition calculator):

http://www.linear-software.com/online.html

This link is provided in the *Appendix B: Resources*.

Following are categorizations of body composition, based on ACSM protocol, as it relates to one's fitness level.

		MALE				
Percentile	Fitness Category	20-29	30-39	40-49	50-59	60 +
90	Well Above Average	7.1-11.7	11.3-15.8	13.6-18	15.3-19.7	15.3-20.7
70	Above Average	11.8-15.8	15.9-18.9	18.1-21	19.8-22.6	20.8-23.4
50	Average	15.9-19.4	19-22.2	21.1-24	22.7-25.6	23.5-26.6
30	Below Average	19.5-25.8	22.3-27.2	24.1-28.8	25.7-30.2	26.7-31.1
10	Well Below Average	25.9 +	27.3 +	28.9 +	30.3 +	31.2 +

		FEMALE				
		20-29	30-39	40-49	50-59	60 +
90	Well Above Average	14.5-18.9	15.5-19.9	18.5-23.4	21.6-26.5	21.1-22.4
70	Above Average	19-22	20-23	23.5-26.3	26.6-30	27.5-30.8
50	Average	22.1-25.3	23.1-26.9	26.4-30	30.1-33.4	30.9-34.2
30	Below Average	25.4-32	27-32.7	30.1-34.9	33.5-37.8	34.3-39.2
10	Well Below Average	32.1 +	32.8 +	35 +	37.9 +	39.3 +

Source: *ACSM's Health Related Physical Fitness Assessment Manual, 2nd Ed. (2008): pg. 59*

Figure 11.7 ACSM body composition chart

Bioelectrical Impedance Assessment (BIA)

The BIA is typically assessed using a handheld unit or scale. During the BIA, a very low electrical current is sent through the body. This current is facilitated by fat-free tissue (i.e., muscle and intercellular water). Fat impedes the electrical current, as there is little water in fat cells. The degree to which the electrical current is impeded is determined and converted to determine body density. That is then converted to percent body fat. These calculations are all done by the BIA unit.

The percent error with BIA is typically associated with one's hydration level. If your client is dehydrated, the result will likely be inaccurate and will show higher levels of body fat than the client actually has (87). Since hydration levels skew the results, it is advisable not to perform the BIA test in the morning or after exercise sessions, as your client will likely be dehydrated. Reassessments using the BIA should be done at the same time of day with the client ingesting approximately the same amount of fluids prior to the assessment. However, because of the possibility of percent error, this testing method is not advised.

Weight

Determining your client's weight as an independent variable is limited in its practicality. By correlating weight and body fat, you will be able to determine what is occurring with body composition. For example, if your client's weight stays the same but body fat decreases, you can deduce that the person gained lean muscle. Many people associate weight as a function of health and fitness. However, it is your responsibility as a coach to educate your clients that it is only one factor and that it needs to be correlated with percent body fat to have real value.

Body Mass Index (BMI)

BMI is calculated using a formula that attempts to determine if an individual is underweight, average weight, or overweight. This is determined by comparing an individual's height and weight. While BMI might be a satisfactory assessment tool for the general population, it is not an accurate tool to use for an athlete, as it does not take body fat or muscle into consideration (88). Therefore someone with a lot of muscle and low body fat would likely be considered "overweight" based solely on BMI. While UESCA does not use BMI as an assessment tool for these reasons, it is important to be aware of what it constitutes and its particular shortcomings when working with athletes.

CLIENT PROFILE

The *client profile* can be completed verbally or via a form sent to a client to fill out, if the person is being coached remotely. The purpose of the client profile is to obtain a wide variety of information to better and more accurately assess/coach an individual. This profile does have a degree of percent error based on the individual as people might answer questions based on what they think they should say rather than reality.

Some of the areas that this profile addresses are:

- Athletic background
- Goals
- Training availability
- Pain tolerance
- Mental readiness to train
- Best type of training structure
- Personality type
- Strengths/weaknesses

UESCA CLIENT PROFILE

1. What type of physical activities do you currently participate in (i.e., sports, weight lifting, etc.)?
2. What competitive sports (if any) have you participated in?
3. Why, when, and how did you get started in running?
4. Will you be training by yourself or with others?
5. What are your short- and long-term running goals?
6. When do you have the most energy, morning or evening?
7. Does your schedule vary or is it fairly consistent?
8. What days/times do you have to train?
9. How important is structure in your life?
10. What aspect of running are you most intimidated by (if any)?
11. Name your top three running-related strengths and weaknesses.
12. Do you respond better to tough love, handholding, or somewhere in between?
13. Do you crack, stay the same, or thrive under pressure?
14. What are you better at? Short, hard sprints and intervals or long endurance efforts?
15. How competitive of a person are you? Rate yourself from 1–10 (10 being the most).
16. How much do you enjoy the intensity of pushing hard in a race/training? 1–10
17. How confident are you in your abilities as a runner? 1–10
18. You like to be challenged 1–10 (1=untrue, 10=very true)
19. Look forward to hard training sessions 1–10 (1=untrue, 10=very true)
20. Susceptible to mental burnout 1–10 (1=untrue, 10=very true)

Figure 11.8 UESCA client profile

❖ PROGRAM DEVELOPMENT VARIABLES

FITNESS LEVEL OF CLIENT

This is a crucial part of developing an effective program. Let us say that you acquire a client who wants to compete in a half marathon in two and a half months but has not started training and has poor aerobic conditioning. What do you do? Do you put the client on a fast-track program that might get the person quasi-ready for the event but would risk injury or do you advise that it would be advantageous to choose another event given the time duration to the event and the current fitness level?

It is important to understand the areas that comprise the fitness component of a training program. Too often, endurance athletes think of “fitness” solely in terms of the cardiopulmonary aspect. The four areas that make up fitness as defined by UESCA are:

1. Cardiopulmonary
2. Strength/Power
3. Stamina/Endurance
4. Flexibility and Joint Range of Motion

All four areas of fitness need to be incorporated into a training program. You likely have heard debates about which sport has the best athletes. Obviously, this is extremely subjective. For example, if people believe that power is the most important trait, they would be hard-pressed to find a sport more targeted than power lifting. However, if one believes that cardiovascular endurance is the most important aspect, running would likely be high on the list. This is noted because this certification is not trying to determine which sport or which aspect of fitness is most important, but rather because it is attempting to identify all areas of fitness that must be included in a program for it to be comprehensive. Each aspect of fitness needs to be scaled properly according to the individual needs of a client. As an example, if your client has a substantially large percentage of body fat, it is advised to focus on fat/weight loss.

The reality is that most running programs are not complete and leave the athlete exposed in certain areas of performance and therefore likely prone to injury.

EXPERIENCE OF CLIENT

Generally speaking, experienced runners present more of a challenge to a coach than beginners, as the areas for improvement aren’t as great. This being said, the principles and fundamentals of coaching a beginner are the same as for an elite athlete. While the focus and specificity are often narrowed for an elite client versus a beginner, the core coaching practices do not change. As a coach, it is important to know how to scale your programming appropriately to accommodate each client.

Experienced Runner

Just because a runner is competing at a high level does not necessarily mean that the person is not deficient in one or more areas.

For many high-level athletes, the role of a coach is not so much to push them, but to provide structure or hold them back when needed if they have the propensity to overtrain.

Experienced runners often challenge coaches because they may have established training methods and may be resistant to change – even if their current training practices are incorrect and inefficient. Being a quality coach means having the ability to communicate effectively with your client as to the *hows* and *whys* of the training process.

If your client is both experienced and elite, the room for improvement is typically small. While the total range for improvement is small, the effort required to coach this type of client is immense. At this level, it's often the small, seemingly trivial insights that make all the difference. Therefore, when working with these clients, your attention to detail must be extremely high. It is also suggested to bring in other professionals (i.e., biomechanists, sport psychologists, etc.) to assist. If you do not feel you can assist certain clients because they are above your level of expertise, you should not work with them and refer them to a more experienced coach.

Novice Runner

When working with a novice runner, and especially one with no history of running, you must be sure to make no assumptions. Assuming a novice client understands and can execute seemingly basic things is a critical mistake.

It is advised that if you are working with novice runners with little to no base fitness, they should be limited to participating in a 10K event during their first year. Additionally, the focus of a first-year runner should be on learning proper mechanics, increasing muscular endurance, gaining cardiovascular fitness, and learning about the sport – not speed. While it is fine to incorporate high levels of intensity into a program, it should be done sparingly and only if the client is physically ready.

TRAINING DURATION

Training duration is the amount of time between the acquisition of a client and the event for which the person is training, assuming the client hired you to help train for a singular event. Before taking on a client, you must take full stock of the following areas to determine if the duration of time to train for the event is enough.

- Client's current training volume
- Experience
- Time availability
- Biomechanical efficiency
- Fitness level
- Distance of event being trained for

Training Volume

The time commitment per week varies greatly based on where the client is in the training process and the event being trained for. For example, at the onset of a training program and during the taper, the weekly time commitment will likely be substantially less than during weeks at peak training volume.

GENETIC PREDISPOSITION

No matter how hard most people work at it, they probably won't be able to dunk like Michael Jordan or swing a club like Tiger Woods. This does not mean, however, that they cannot improve substantially. You might know of runners who can go out and run 10 miles without much training, whereas most others would be exhausted after just one mile. Whether it is muscular endurance, VO_2 max, ability to recover, or pain tolerance, your client will likely find certain areas of training and development easier than others.

If your client is experienced, the person likely will have a good idea of where his or her inherent strengths and weaknesses lie. However, if your client is new to the sport or to physical activity in general, it will take some time for you to be able to determine strengths. Once you ascertain this, it will help with program development, as you will be able to target areas that likely need more focus than others.

The Sport Gene by David Epstein examines the role of nature (genetics) versus nurture (training) in regard to sports performance (466). While the book is full of research and interesting anecdotes regarding this topic, the gist of the book is that both genetics and training impact the success rate of an individual at a particular sport. Genetics influence many things, including how well an individual adapts to aerobic exercise, limb lengths, height, etc. Based on the sport, a genetic attribute such as extreme height might be an asset (basketball) whereas in another sport it is largely a liability (distance runner) (466).

As noted in the book, in a study, 1,900 Canadian firefighter applicants were given VO_2 max tests to see if there was such a thing as a naturally gifted aerobic individual. As it turns out, the study identified six individuals who had no history of aerobic training but had VO_2 max levels associated with collegiate distance runners (479).

Genetics may also play a role in how motivated your client is to exercise and train. A study of 37,051 twins found that 48 to 71 percent of the variation of the amount of exercise performed by the subjects could be attributed to genetics (495).

From a biomechanical standpoint, there are traits that correspond to being an efficient runner. For example, a small calf girth is an advantage for runners as the lower leg acts as a pendulum.

WORK ETHIC OF CLIENT

Some clients will have a stronger work ethic than others. Some clients will hang on your every word and will do anything and everything you say. For other clients, it will be like pulling teeth to get them to come close to adhering to the training program. The majority of clients will fall somewhere between these two extremes.

The stronger your client's work ethic is, the faster the person will typically progress and see results. While one aspect of being a coach is the ability to motivate, it does not always ensure that your client will give 100 percent. If the work ethic of your client is poor, it is your responsibility to try to find a way to motivate the individual. If the training program is modified to accommodate a client's lack of training, you must reschedule the goal event if the training will not adequately prepare the client for the event.

You must be honest in your assessments of your clients. This might call for tough conversations when addressing a client's lack of work ethic and adherence to the program. While being tactful in your conversations, you must be direct and truthful concerning current training performance.

TRAINABILITY

This relates to how well an individual responds to the training process. Also in *The Sport Gene*, Epstein notes a study called the HERITAGE Family Study (466). The general thesis of this study is that individuals respond differently to training, largely based on genetic factors (467, 468). Specifically, in 2011 HERITAGE researchers identified certain gene variants associated with high aerobic capacity. Individuals who had a large number of these gene variants improved their VO_2 max three times more than those who had a low number. Other gene variants correlated with the magnitude drop in heart rate due to aerobic training (466, 467, 468).

Trainability relates to training methodology. In other words, if a client is not responding well to a particular training method or application, change the training stimulus (466). A training method or application must be implemented for a long enough period to ensure that a training adaptation had enough time to occur.

SPECIAL POPULATIONS

16 YEARS OLD OR YOUNGER

Below are five overriding themes when coaching any youth sport:

1. Fun
2. Sportsmanship
3. Skill and motor development

4. Making friends
5. Safety

This is not to say that these things should not be at the forefront of an adult's program, but they definitely need to be the guiding principles when coaching youth. Remember, kids are not mini-adults, they're kids!

Involving Parents

It is a parent's child, not yours. Therefore, you need to include the parents in the training process – as long as their participation is constructive and supportive. The exact verbiage of how you speak to a child's parents is up to you. The primary things to remember when involving parents are:

- Keep them in the training process
- Keep lines of communication open
- Be supportive of their goals for their child as long as they are healthy and attainable
- Manage their expectations
- Continue to reiterate your training philosophy throughout the coaching process
- Be respectful and professional at all times

It is important to understand that you are coaching a child who, more than likely, is not just participating in running events. The child is most likely doing all sorts of sports and nonsport-related activities. Therefore, you should look at the training process as an opportunity to enhance the young client's motor development. This means you should work in all planes of movement and engage the child in movements and activities that may not necessarily be running-specific.

Communication

How you communicate with a child is different from how you should communicate with an adult. A youth participant is less likely to be interested in the nuances of the training process than an adult would be. Keeping the science out of the verbiage when coaching a youth client is highly advised.

It is probable that a youth client won't have a long attention span; the younger the client, the more likely this is. Therefore, save your 20-minute dissertation on oxidative phosphorylation for your adult clients and try to keep talks with youth clients to a few minutes and on topics they understand.

The emphasis of communication should be lighthearted, fun, and supportive. Keep criticism to a minimum, and if criticism is used, it must always be framed in a supportive manner. Foul language should never be used when communicating with clients, regardless of age.

Safety

When training youth (and adult) clients, the most important aspect is safety. The musculoskeletal structure of a youth athlete is still in a developmental phase and cannot and should not have the same stressors placed upon it as an adult. The younger the athlete is, the more this is true. Areas where this applies most are race distance, strength training, and race/training frequency and intensity.

Regarding distance, it is the position of UESCA that athletes 12 years old and younger should be limited to participating in events 10K and shorter.

While there are no national standards or rules in regard to what age a child should be to participate in running events, as it pertains to longer events such as half and full marathons, below are four major running events in regard to their respective age requirements:

- **Rock 'n' Roll Marathon Series:**
  - Must be 12 to participate in half marathons
  - Must be 18 to participate in full marathons

- **Walt Disney World Half Marathon:**
  - Must be 14 to participate in the half marathon
    (For 10K, must be at least 10 years old)

- **NY Marathon, Boston Marathon:**
  - Must be 18 years old to participate in full marathon

Adolescents mature at very different rates, and as such, it is difficult to determine if an adolescent is physically mature enough to handle a set distance. Therefore when working with youth runners, a slow and steady progression is always the best course of action as you want to ensure the running program is properly prescribed and scaled.

The focus on strength training should be on body weight exercises in all planes of movement, with the primary focus on overall strengthening and fun.

Fun

Running should be fun for a child. If running is something a parent wants for a child but the child hates it, this is a time when a conversation between you and the parent is appropriate. How you decide to make training fun is up to you, but you should try your hardest to strike a balance between fun and some form of structured training.

Youth under the age of 14 should have a minimal amount of specialization in the program.

Physical Development

The physical development of children largely relates to when they hit puberty. Kids hit puberty at different times. For this reason, assessing the future potential of a child is very difficult as one's success may have more to do with hitting puberty early than it does with being a running prodigy.

While a physically mature child might be able to handle a slightly higher workload than a child who has not yet hit puberty, this should be assessed on an individual basis. It is never a wrong decision to progress a child conservatively.

65 YEARS OLD OR OLDER

It is a fact that as one ages many physiological changes occur. Following are some examples of these changes (92):

- Cardiac output decreases
- Overall decrease in elasticity of muscles and connective tissue
- Decrease in nerve impulses (e.g., reaction time)
- Sarcopenia: a decrease in muscle strength
- Decrease in lung capacity
- Bones become weaker and more brittle (i.e., osteoporosis)
- Increase in recovery time

Does this mean that if you are coaching an athlete who is 65 or older there is no hope? Absolutely not, and in fact, it is quite the opposite. As a coach of an older adult, you have the opportunity to have a substantial impact on slowing the aging process. As an example of what fitness can do to slow the aging process, at the 2015 Fifth Avenue Mile in NYC, the winner of the 70- to 74-year-old age group crossed the line in 6:11 and the winner of the 80–99 age group did it in 7:00 (93)!

A comprehensive training program can dramatically slow physiological changes associated with aging, especially muscle strength (639). Some important areas to identify in a training program are:

1. Strength training
2. Adequate recovery periods
3. Maintaining intensity in the training program

The overall structure of a running program for clients 65 years and older should be based on the individual, not the age. This is because of the wide range of physical capacity.

Muscular Endurance

While muscle strength and power have been shown to decrease over 60 years of age (640), studies seem to show that muscular endurance is not affected (641). The neuromuscular system in regard to running is largely correlated to muscle fatigability. In other words, as someone ages, the fatigability does not change much, if at all.

This is why you often hear of professional runners moving from shorter distances to longer disciplines (i.e., marathon) as they get older. This is because long-distance events place less emphasis on muscle power and more on muscular endurance.

Areas of Focus

While muscular endurance has not been shown to decrease with age, as noted previously, the same cannot be said for muscular power/strength, cardiovascular output, lung capacity, and reaction time. Additionally, posture and gait control have been shown to decrease with age (642).

Therefore when creating a training program for a runner over 65 years old, focusing on these aforementioned areas in a safe manner is advised. Of course, age is a relative number and while it is likely that you will come across a runner in the 40s who requires much more attention than a runner in the 60s, the general trend is that older runners should have different areas of focus than younger runners.

CLIENTS WITH A PHYSICAL DISABILITY

Training methods differ based on the physical limitations of an athlete. Regardless of the disability, you need to focus on the same areas of performance enhancement and components of fitness that you would with an able-bodied individual. It is how you customize the program and workouts to accommodate a client that will differ.

Important areas on which to focus when working with a physically challenged client are:

- Safety
- Feedback
- Equipment

Depending on the particular area of disability, safety and equipment might be more of an issue than with an able-bodied client.

According to Catherine Sellers, director of *USA Paralympic Track and Field High Performance*, a coach goes through a growth process when first working with an athlete with a physical disability (342). The stages are as follows:

1. Fear of the unknown
2. Fear of saying something wrong
3. Shock
4. Acceptance
5. Advocacy

When working with any athlete, the learning process is a two-way street. This is especially true when working with a physically disabled athlete. While the athlete is learning about training methodologies, physiology, and race tactics from you,

you are learning about the person's particular disability and the challenges that it presents.

As a coach, one of the most important and necessary traits is the ability to adapt. As stated above, the goal and basic structure of the program should not change; it is just a function of what modifications need to be implemented to meet the needs of a client.

If your client has a video of him or herself racing or videos of others with the same disability, it can be very helpful in understanding the nature of the disability and its impact on performance.

❖ SUMMARY

- An *HHQ* and *PARQ* must be completed before training a client can commence.
    - If a client answers *yes* to any of the PARQ questions, the person *must* get a physician's clearance prior to working with you.
- Based on UESCA protocol, any client, regardless of PARQ or HHQ results, *must* have been cleared by a physician in the last 12 months prior to beginning a running program.
- It is advised that your client receive a cardiac assessment within one year prior to beginning a training/racing program.
- A *Gulick tape measure* is used to take accurate girth measurements.
- It is advised not to use BMI or BIA methods to determine body composition because of a substantial chance for inaccuracies.
- Many variables come into play when creating a training program. Below are six variables:
    - Fitness level
    - Experience level
    - Training duration (how long before goal event)
    - Genetic predisposition
    - Work ethic of client
    - Special populations
- The client profile will help you to assess clients in the following areas:
    - Athletic background
    - Goals
    - Training availability
    - Pain threshold
    - Mental readiness to train
    - Best type of training structure
    - Personality type
    - Strengths/weaknesses
- Fun, safety, and age-appropriate communication are focal points when coaching youth clients.
- When working with youth clients, parents must be involved in the training process.
- While physiological capacity decreases as people age, physical training can dramatically slow the aging process.

- Clients with a physical disability present a unique set of challenges to a coach, primarily regarding the use of adaptive equipment. In these cases, a coach has an opportunity to learn as much as a client.